

Morphometry of terrestrial shield volcanoes

Pablo Grosse^{a,b,c,*}, Matthieu Kervyn^c

^a Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET), Argentina

^b Fundación Miguel Lillo, Miguel Lillo 251, 4000 San Miguel de Tucumán, Argentina

^c Department of Geography, Earth System Science, Vrije Universiteit Brussel, Pleinlaan 2, B-1050, Belgium

ARTICLE INFO

Article history:

Received 21 February 2017

Received in revised form 5 October 2017

Accepted 11 December 2017

Available online 14 December 2017

Keywords:

Shield volcanoes

Shield-like volcanoes

Morphometry

Hierarchical cluster analysis

ABSTRACT

Shield volcanoes are described as low-angle edifices built primarily by the accumulation of successive lava flows. This generic view of shield volcano morphology is based on a limited number of monogenetic shields from Iceland and Mexico, and a small set of large oceanic islands (Hawaii, Galápagos). Here, the morphometry of 158 monogenetic and polygenetic shield volcanoes is analyzed quantitatively from 90-meter resolution SRTM DEMs using the MORVOLC algorithm. An additional set of 24 lava-dominated 'shield-like' volcanoes, considered so far as stratovolcanoes, are documented for comparison. Results show that there is a large variation in shield size (volumes from 0.1 to > 1000 km³), profile shape (height/basal width (H/W_B) ratios mostly from 0.01 to 0.1), flank slope gradients (average slopes mostly from 1° to 15°), elongation and summit truncation. Although there is no clear-cut morphometric difference between shield volcanoes and stratovolcanoes, an approximate threshold can be drawn at 12° average slope and 0.10 H/W_B ratio. Principal component analysis of the obtained database enables to identify four key morphometric descriptors: size, steepness, plan shape and truncation. Hierarchical cluster analysis of these descriptors results in 12 end-member shield types, with intermediate cases defining a continuum of morphologies. The shield types can be linked in terms of growth stages and shape evolution, related to (1) magma composition and rheology, effusion rate and lava/pyroclast ratio, which will condition edifice steepness; (2) spatial distribution of vents, in turn related to the magmatic feeding system and the tectonic framework, which will control edifice plan shape; and (3) caldera formation, which will condition edifice truncation.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

Shield volcanoes have long been recognized as a specific type of volcano (e.g. Cotton, 1944; Whitford-Stark, 1975), although its definition remains vague. It generally refers to monogenetic or polygenetic volcanic constructs with low slopes built up primarily by the accumulation of low-viscosity basaltic lava flows (e.g. Macdonald, 1972; Walker, 2000; Werner, 2014). Shield volcanoes are common in several tectonic settings, mainly as large hotspot-induced oceanic island volcanoes, but also as fields of monogenetic to small polygenetic edifices in tectonic rift (e.g. Iceland - Rossi, 1996) or back-arc settings (e.g. Mexico - Hasenaka, 1994). Shields are also very common on other planets where their morphology is preserved from erosion (Pike, 1978; Kreslavsky and Head, 1999; Plescia, 2004; Spudis et al., 2013).

Previously proposed classifications of volcano landforms have mixed morphology with other criteria including magma composition and types of volcanic products (e.g. Macdonald, 1972; Pike, 1978; Pike and

Clow, 1981; Davidson and De Silva, 2000; Francis and Oppenheimer, 2003; Siebert et al., 2010). In these classifications, 'shield volcanoes' are always a recognized volcano type, but no clear-cut quantitative definitions are offered and in most cases no quantitative sub-divisions are attempted. Whitford-Stark (1975) and Pike (1978) did classify shields into sub-types based on morphology, but they focused mostly on volcanoes from the continental USA, Hawaii, Galápagos and Iceland, for which topographic maps were then available. The recent availability of global topographic datasets (e.g. the SRTM DEMs) offers the opportunity to systematically document the morphological variation of shield volcanoes worldwide. Morphometric analysis has the potential to quantitatively compare volcano morphologies, to identify size-independent morphological similarities, and to isolate the controlling factors (e.g. Grosse et al., 2009, 2014).

We here apply the MORVOLC algorithm (Grosse et al., 2012, 2014) to systematically document the shape and size of 182 shield and shield-like volcanoes from contrasted tectonic settings around the world. Applying principal component and cluster analyses, we identify the key parameters that characterize their morphometry and end-member morphological types. We finally propose a model that integrates morphologies into evolutionary pathways of shield volcano growth highlighting the key factors controlling these evolutions.

* Corresponding author at: Fundación Miguel Lillo, Miguel Lillo 251, 4000 San Miguel de Tucumán, Argentina.

E-mail address: pgrosse@lillo.org.ar (P. Grosse).

2. Review of shield volcano morphology

Most volcano textbooks define shield volcanoes with three main characteristics: (1) gentle slopes; (2) primarily or entirely composed of lava flows; and (3) basaltic magma composition (e.g. Walker, 2000; Francis and Oppenheimer, 2003; Lockwood and Hazlett, 2010), although shields with other compositions have also been documented, such as andesitic (e.g. Hasenaka, 1994) or trachytic (Webb and Weaver, 1975). Lava flows building shields are seen as low-viscosity lavas forming thin and long flows. The characteristic resulting shape is that of an upward-convex topographic profile, attributed to the thickening of lava flows with distance and the occurrence of pit craters or collapse calderas at the summit (Francis and Oppenheimer, 2003). Average slopes are generally mentioned to vary between 2 and 10° (Macdonald, 1972), <10° (Cas and Wright, 1988), 4 and 8° (Walker, 2000), 5 and 10° (Zimbelman, 2000; Francis and Oppenheimer, 2003), or <15° (Lockwood and Hazlett, 2010), but specific studies have shown that steeper slope angles can be identified (Mouginis-Mark et al., 1996; Rowland and Garbeil, 2000).

The first classification of shields was proposed by Whitford-Stark (1975), who discriminated five types based mostly on order-of-magnitude size variations: scutulum, Icelandic, Galápagos, Hawaiian and macro-shields of Mars. Although size was not retained as the most important parameter, subsequent classification schemes kept the distinction of Icelandic, Galápagos and Hawaiian shields as main shield types. Pike (1978) and Pike and Clow (1981) proposed a new classification with five shield classes: large shields with a caldera of either tholeiitic or alkali basalt composition, and smaller shields with a crater, within which Icelandic, low or steep slope types were discriminated. Wood (1979) showed that small shields (termed 'lava cones' in Wood, 1979) are clearly discriminated morphologically from pyroclastic cones, but that the distinction between the three types of small shields of Pike (1978) was poorly defined and mostly characterized by contrasted size ranges.

Probably the most used volcano classification is that of the Smithsonian Institution's Global Volcanism Program (GVP) compilation of active and potentially active volcanoes of the world (Siebert et al., 2010). Of the >1500 listed volcanoes, 170 are classified as shields (and an additional 8 are considered as 'pyroclastic shields', i.e. mainly composed of pyroclastic deposits), although the classification is not based on any systematic morphometric analysis, but rather on the compilation of existing references, which in turn usually rely on qualitative morphology and/or eruption style and composition to define a shield volcano as such.

Small monogenetic single-vent shield volcanoes are usually discriminated from large polygenetic shields characterized by numerous vents on the edifice summit and flanks. Monogenetic shield volcanoes typically consist of the accumulation of lava flows around a central vent forming a proximal cone surrounded by a broader and flatter lava apron (Hasenaka, 1994; Rossi, 1996). They are usually characterized by a flat summit plateau or a crater, representing the location of a lava lake feeding the lava overflows that build the shield (Rossi, 1996). The morphometry of small shields has been studied in Central Mexico (Hasenaka, 1994) and Iceland (Rossi, 1996; Pedersen and Grosse, 2014). Hasenaka (1994) discriminated two types of shields, with lower (~5°, A-type) and steeper (~10°, B-type) slope angles. In Iceland, Rossi (1996) also discriminated two types of shields associated with different average slopes and magma composition: olivine tholeiite shields with 3.4° average slopes and picrite basaltic shields with 5.8° average slopes. Another type of monogenetic shield is the extremely flat constructs (1–2° slopes) associated to basaltic plains volcanism, such as those of the Snake River Plain (Idaho, USA; e.g. Greeley, 1982; Kuntz et al., 1992).

The morphology of large polygenetic oceanic island shields has received more attention. Specific studies have focused on the analysis of a single or a group of shields, e.g. Galápagos islands (Nordlie, 1973;

Cullen et al., 1987; Rowland et al., 1994; Rowland, 1996; Naumann and Geist, 2000), Hawaiian islands (Moore and Mark, 1992; Bleacher and Greeley, 2008), Piton de la Fournaise, Réunion island (Michon and Saint-Ange, 2008). Studies of multiple oceanic shields have especially highlighted the variation of slope angles between shields and at different elevations within each edifice (Mouginis-Mark et al., 1996; Rowland and Garbeil, 2000). Most large shields are characterized by simple or complex summit caldera structures, but also by broad flat summit areas or plateaus that have been attributed to caldera infilling or to deformation of the hydrothermal system of the volcano (Merle et al., 2010). Large polygenetic shield or shield-like (in the sense of Davidson and De Silva, 2000) volcanoes also occur in continental settings. Examples are Mount Cameroon (Kervyn et al., 2014), several shields in the East African Rift System (e.g. Barberi and Varet, 1970; Webb and Weaver, 1975), and shield or shield-like volcanoes in arc (e.g. Westdahl, Miller et al., 1998) or back-arc (e.g. Newberry, Higgins, 1973; Medicine Lake, Donnelly-Nolan et al., 2008; Payún Matrú, Hernando et al., 2012) settings.

According to Francis and Oppenheimer (2003), the slope profile of shields is entirely controlled by the rheology of the lava they are made of. Several factors have however been proposed to contribute to the steeper-than-expected slopes of many small and large shields. The most common factors include: (1) higher viscosity lavas, associated with different chemical compositions (Hasenaka, 1994; Rossi, 1996; Wolfe et al., 1997); (2) lower effusion rates producing shorter and thicker flows (Chadwick and Howard, 1991; Hasenaka, 1994); (3) contrasted lava flow types with distinct length/thickness ratios erupted from different vents (Chadwick and Howard, 1991; Rossi, 1996); (4) accumulation of pyroclastic material along preferential zones (Hasenaka, 1994; Rowland and Garbeil, 2000); and (5) preferential dyke intrusion along rift zones or circumferential fissures (Chadwick and Dietrich, 1995; Annen et al., 2001). These different factors are related to each other: the distribution of dyke intrusions, resulting from the local stress field, will control the spatial distribution of volcanic vents that, in turn, will control the location of pyroclastic material accumulation and the source of contrasted lava flow types (Rowland and Garbeil, 2000; Tibaldi et al., 2014).

Large shield volcanoes on Earth and on Mars are also affected by gravitational deformation including rapid mass wasting (Michon and Saint-Ange, 2008; Michon et al., 2009), slow flank slumping (Morgan et al., 2003), volcano-scale sagging caused by lithosphere flexure (Byrne et al., 2013, 2015) or volcano spreading (Platz et al., 2011; Kervyn et al., 2014). Such volcano-scale deformations generate locally steep slopes or modify the entire flank profile, and influence the local stress field and therefore the distribution of volcanic vents (Kervyn et al., 2009; Tibaldi et al., 2014).

3. Methodology

3.1. Data sources

The main data source used for the morphometric analysis of the shield volcanoes was the near-global coverage, C-band 3 arc-seconds (~90-m spatial resolution) Shuttle Radar Topography Mission (SRTM) digital elevation model (DEM) (e.g. Rabus et al., 2003). The seamless dataset from CGIAR-CSI (Consultative Group on International Agricultural Research–Consortium for Spatial Information; Jarvis et al., 2008) was used. Additionally, for volcanoes of Iceland that are not covered by the SRTM DEM, a 90-m resolution DEM derived from photogrammetry of aerial images was used.

3.2. Selection of volcanoes

The selection of volcanoes for analysis was based on the GVP database (Siebert et al., 2010). The database lists 178 shield volcanoes (including 8 'pyroclastic shields'). Most of these are large, polygenetic

edifices, but several small, monogenetic shields (mainly from Kamchatka) are also listed individually. In addition, several entries listed as ‘volcanic fields’ or ‘cinder cones’ contain numerous relatively small, monogenetic shield volcanoes. Although not included in the GVP database, we also considered shield volcanoes from Iceland because they have pristine morphologies and are an established shield type. From all these potential entries, we selected only volcanoes with: (1) well-preserved constructional morphology, with no or minor erosive features; (2) well-defined boundaries, with no or limited overlap with surrounding topography; (3) average basal width larger than 3 km (arbitrary limit established to discard edifices that are too small to be accurately analyzed with the 90-m DEM resolution). In addition, volcanoes from Alaska above 60°N latitude and from Antarctica, not covered by the SRTM DEM, had to be excluded. We also discarded Pleistocene shields with recent activity restricted to non-shield volcanism (e.g. domes, cinder cones); for example, the Pinacate volcanic field (NW Mexico) consists of an older Lower Pleistocene shield volcano (Santa Clara) covered by hundreds of late Pleistocene to Holocene cinder cones and maars (e.g. Gutmann, 2007). Following these criteria, a total of 158 shield volcanoes were selected for analysis. In addition, we also included in our analysis 24 volcanoes listed in the GVP database as stratovolcanoes but that are low, shield-like, lava-dominated edifices. The 182 analyzed volcanoes are listed in Electronic supplement Table 1 and their worldwide distribution is shown in Fig. 1. Of these, 103 were previously considered in the global database of Grosse et al. (2014).

3.3. Volcano outline delineation

The basal outlines of the shield volcanoes were defined considering concave breaks-in-slope around the edifice base. Flat-lying aprons and other far-reaching volcanic products are excluded from the analysis. For volcanic islands where the edifice continues underwater, only the subaerial part of the volcano is considered. The basal outlines of the 103 volcanoes previously analyzed by Grosse et al. (2014) were determined using the NETVOLC algorithm (Euillades et al., 2013). For the present study, the same NETVOLC outlines were maintained for 80

edifices, whereas for the other 23 volcanoes minor adjustments were manually performed. For the 79 shields not covered by Grosse et al. (2014), manually determined basal outlines, following the same approach, were used. For details on the issue of volcano boundary delineation see Grosse et al. (2012, 2014) and Euillades et al. (2013).

In addition to the basal outlines, 40 summit caldera outlines, previously obtained manually by Grosse et al. (2014), were also taken into account. We considered only well-defined calderas with average diameters > 1.6 km (the traditionally considered minimum size limit of calderas, e.g. Macdonald, 1972).

3.4. Computation of morphometric parameters with the MORVOLC algorithm

Morphometric parameters for each volcano edifice were extracted from the DEMs using the MORVOLC algorithm (Grosse et al., 2009, 2012, 2014). The obtained parameters characterize the morphometry of each edifice in terms of size (basal and summit areas (A_B , A_S) and widths (W_B , W_S), height (H , H_{MAX}), volume (V , V_{MAX})), profile shape (height/basal width (H/W_B) and summit width/basal width (W_S/W_B) ratios), plan shape (ellipticity and irregularity indexes of both the basal outline ($El_{OUTLINE}$, $Il_{OUTLINE}$) and averages of the main flank elevation contours of the edifice (El_{AVG} , Il_{AVG})), slopes (mean slopes of the whole edifice (S_{TOT}), of the lower flank (S_{LFL}), of the main flank (S_{MFL}), and of the summit region (S_{SR})), the number of secondary peaks (PK) and peak density ($PK_{DENSITY}$). Additional parameters were obtained for the delineated summit calderas. Fig. 2 is a sketch showing the main calculated parameters and Electronic supplement Table 2 describes all of the parameters used. Further details on the parameters can be found in Grosse et al. (2012, 2014).

3.5. Principal component and cluster analyses

Principal component and cluster analyses were applied to the morphometric database. For these statistical analyses, a subset of 130 volcanoes (107 shield volcanoes and 23 shield-like stratovolcanoes) and 19

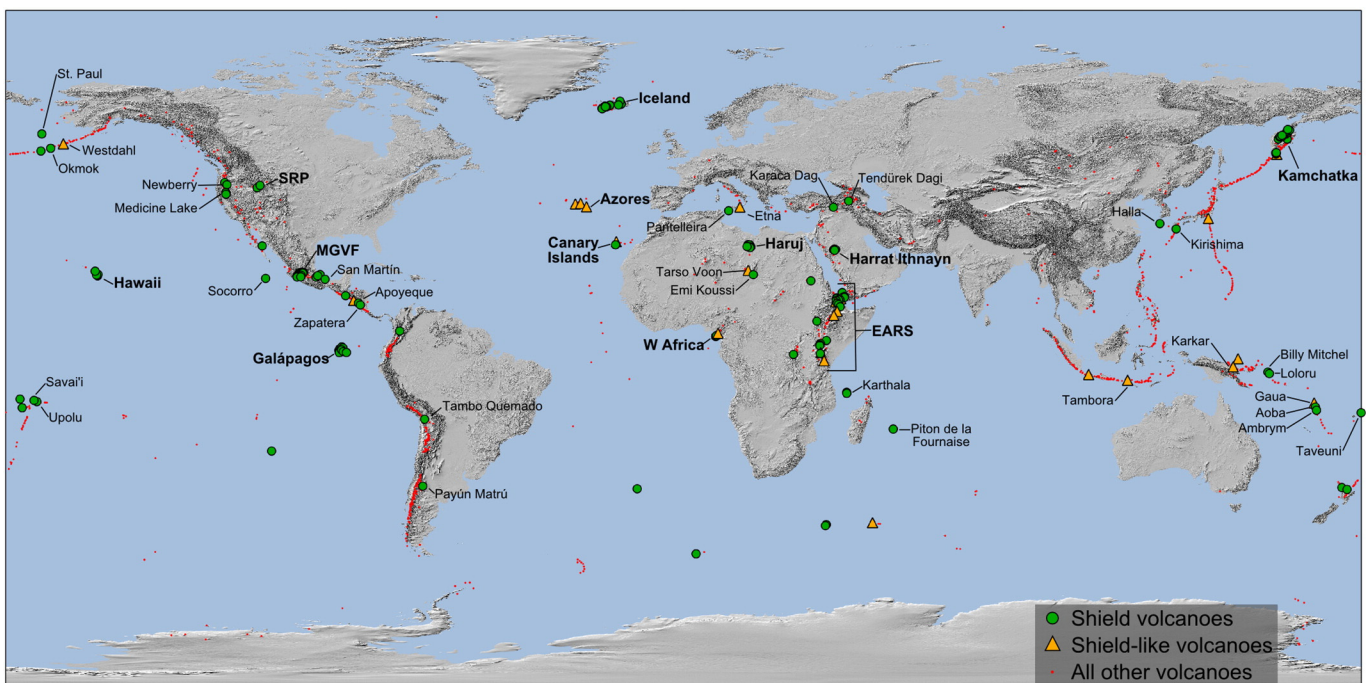


Fig. 1. World map showing location of 158 shield and 24 shield-like volcanoes considered in this study. All other volcanoes listed in the Smithsonian Institution's Global Volcanism Program are also shown. Individual shield volcanoes and regions mentioned in the text are indicated. EARS: East African Rift System; MGVF: Michoacán-Guanajuato Volcanic Field; SRP: Snake River Plain.

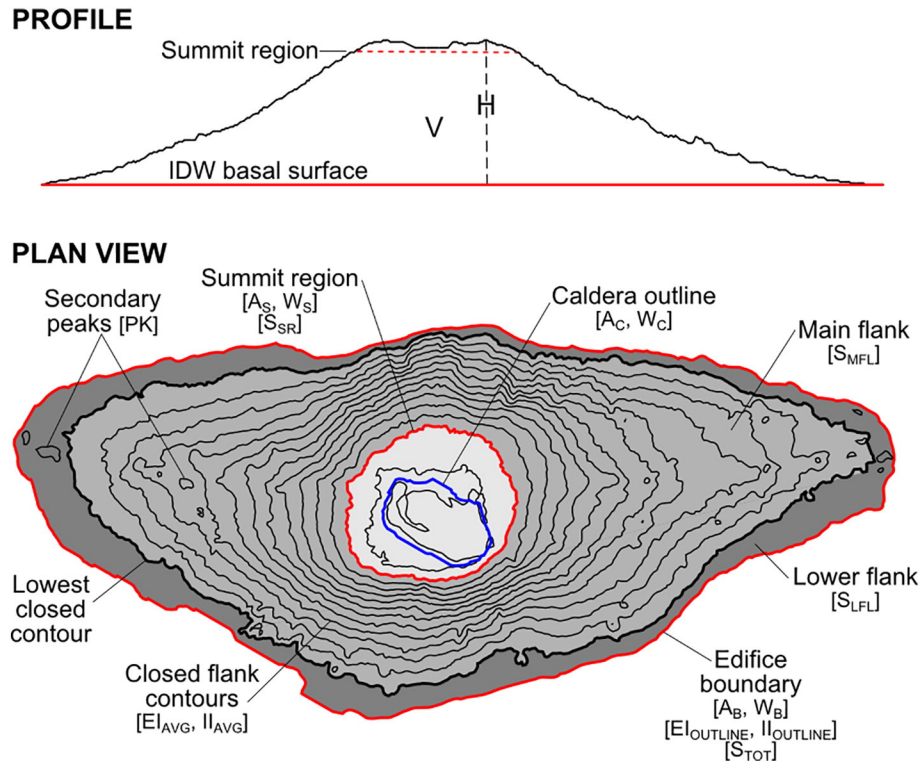


Fig. 2. Schematic profile and plan view of a shield volcano showing main computed morphometric parameters: height (H), volume (V), basal area and width (A_B , W_B), summit region area and width (A_S , W_S), caldera area and width (A_C , W_C), ellipticity and irregularity indexes of the basal outline ($EI_{OUTLINE}$, $II_{OUTLINE}$) and averages of the flank elevation contours (EI_{AVG} , II_{AVG}), mean average slopes (S_{TOT} : of the whole edifice, S_{LFL} : of the lower flank, S_{MFL} : of the main flank, S_{SR} : of the summit region), and number of secondary peaks (PK). IDW: inverse distance weighting. For description of parameters see Electronic supplement Table 2.

parameters were considered, discarding volcanoes with few and/or inaccurate elevation contours in terms of definition of summit region and of EI_{AVG} and II_{AVG} , and focusing on the morphometric parameters available for all studied volcanoes. A principal component analysis (PCA) linearly transforms data into a new set of uncorrelated principal components sorted according to the variance explained in the original data (Jolliffe, 2002). Each principal component accounts for part of the variability of several original parameters, according to their loading on each component. In our case, the first four principal components, which explain 85% of the variance, were then used as input for hierarchical cluster analysis (e.g. Köhn and Hubert, 2015). Cluster analysis refers to a set of statistical techniques that aim at grouping entries of a dataset into several clusters, in such a way that entries in one group are more similar to each other than those in other groups. We applied three hierarchical clustering methods: farthest neighbour or complete linkage (where the distance between two clusters is defined as the largest distance between items), average linkage or UPGMA (unweighted pair group method with arithmetic mean, where the distance between two clusters is the average distance for all pairs of items between each cluster, weighted by the number of items in each cluster), and weighted centroid (where the distance between two clusters is defined as the distance between the centroids of each cluster). Both PCA and cluster analyses were conducted with the ENVI/IDL software package.

4. Results: morphometry of shield volcanoes

The analyzed volcanoes cover a wide spectrum in terms of geographical location, tectonic setting and composition (Electronic supplement Table 1; Fig. 1). Following the GVP database, 25 volcanoes are in continental rift zones, 39 are in oceanic rift zones, 68 are in continental subduction zones (including back-arc volcanism), 12 are in oceanic subduction zones, 28 are in continental intraplate settings and 10 are in oceanic intraplate settings. Most (114) are of dominant basaltic

composition, but others are dominantly andesitic (37), trachybasaltic (12), trachytic/trachyandesitic (9), rhyolitic (7), phonolitic (2) or dacitic (1). Many have secondary vents and/or cinder cones on their flanks, ranging in number from a few to hundreds. 40 have large well-defined summit calderas, and several others have filled calderas or caldera remnants. Many others have smaller summit craters.

The obtained morphometric data is presented in Electronic supplement Table 1. Table 1 summarizes the morphometry of the analyzed shield and shield-like volcanoes; the statistics for all other non-shield volcanoes (i.e. stratovolcanoes, complex/compound volcanoes) in the database of Grosse et al. (2014) are also shown for comparison. Fig. 3 shows histograms for volume, the H/W_B ratio and mean total slopes (S_{TOT}), and several scatter plots showing the ranges of, and correlations between, some of the morphometric parameters. The size of shields spans the complete spectrum for volcanoes, and most very large volcanoes (basal widths > 25 km; volumes > 200 km³) are shield or shield-like (Fig. 3a, d, e). Volumes range from 0.1 to 1000 km³, except for Mauna Loa and Kilimanjaro, each with volumes of 3000 km³ (note that all volumes are above sea level only). Excluding these outliers, shield and shield-like volcanoes have a mean volume of 78 km³ and a median of 7.4 km³. The volume distribution is log-normal and bimodal, with peaks at ~ 1 –3 km³ and ~ 50 –180 km³ and a lower number of shields between ~ 20 –50 km³ (Fig. 3a). This volume range with few shields can be used to distinguish between small and large shields, corresponding approximately to the limit between the ‘Icelandic-type’ and ‘Galápagos-type’ shields of Whitford-Stark (1975).

As expected, average slopes and H/W_B ratios are considerably lower for shields than for non-shields, but there is substantial overlap. 90–95% of shields have $S_{TOT} < 12^\circ$ and H/W_B ratios < 0.10 (Fig. 3b, c, f, g, h), with overall median values of 6.4° and 0.05, respectively. In comparison, the median values for non-shield volcanoes are 17.5° and 0.13. The shield-like volcanoes have S_{TOT} and H/W_B ratios within the range of shields (Table 1; Fig. 3b, c, f, g, h), although their median values are somewhat

Table 1

Summary statistics of main morphometric parameters for shield volcanoes, shield-like stratovolcanoes and all other stratovolcanoes/complex volcanoes. (From the database of Grosse et al. (2014).)

	Basal width	Height	Volume	Height/basal width	Summit width/basal width	Ellipticity index [avg. contours]	Irregularity index [avg. contours]	Mean slope [total]	Peak number	Peak density	Calderas
	W_B (km)	H (km)	V (km^3)	H/ W_B	W_S/W_B	El_{AVG}	Ir_{AVG}	S_{TOT} (°)	PK	$PK_{DENSITY}$ ($1/\text{km}^2$)	Num. (%)
											Width (km)
Shield volcanoes (N = 158)											28 (18%)
Median	7.6	0.46	4.2	0.050	0.16	1.49	1.11	6.4	3	0.06	4.6
Mean	13	0.64	90	0.053	0.20	1.74	1.17	7.1	12	0.08	5.1
St. dev.	12	0.60	290	0.028	0.13	0.68	0.18	3.9	25	0.07	3.1
Minimum	2.6	0.03	0.14	0.006	0.01	1.12	1.01	1.1	1	0.00	1.6
Maximum	69	3.03	3031	0.179	0.64	5.23	2.25	21.0	166	0.43	12.3
Shield-like volcanoes (N = 24)											12 (50%)
Median	15	0.81	47	0.070	0.28	1.55	1.18	9.8	10	0.05	5.2
Mean	18	1.28	244	0.071	0.31	1.70	1.23	10.0	19	0.06	5.8
St. dev.	14	1.04	628	0.022	0.16	0.63	0.13	2.2	30	0.04	4.0
Minimum	4.3	0.25	1.2	0.023	0.03	1.24	1.07	7.1	1	0.01	1.8
Maximum	66	4.57	3010	0.116	0.63	4.46	1.51	16.9	143	0.18	16.3
Non-shield volcanoes (N = 647)											33 (5%)
Median	8.4	1.05	15	0.129	0.18	1.70	1.18	17.5	5	0.11	2.3
Mean	9.6	1.12	37	0.133	0.21	1.80	1.30	17.7	11	0.15	3.4
St. dev.	5.6	0.51	65	0.049	0.15	0.54	0.37	4.1	19	0.13	2.5
Minimum	2.0	0.17	0.27	0.036	0.02	1.09	1.01	6.4	1	0.01	1.6
Maximum	38	3.09	780	0.300	0.76	4.35	3.73	33.3	234	1.38	11.9

For description of parameters see Electronic supplement Table 2.

higher (9.8° and 0.07 , respectively). These higher average values can be related, at least in part, to the lack of very small and very flat shield-like volcanoes in our analysis (none of the 24 considered shield-like volcanoes have $S_{TOT} < 7^\circ$ and only one has a volume $< 5 \text{ km}^3$) as opposed to an abundance of this type of shield (Fig. 3f, g). Shield and shield-like volcanoes with dominant basaltic compositions (according to the GVP database) have lower S_{TOT} and H/W_B ratios (median values of 6.0° and 0.04 , respectively) compared to those of other dominant compositions (median values of 9.1° and 0.07 , respectively) (Fig. 3b, c), and 43 of the 49 (88%) shields with $S_{TOT} < 5^\circ$ are basaltic. Interestingly, this compositional difference does not apply to the non-shield volcanoes, as the median values of S_{TOT} (17.7°) and H/W_B ratios (0.14) for the basaltic non-shields are similar to those of the non-basaltic non-shields (17.4° and 0.13 , respectively) (Fig. 3b, c).

Edifice truncation, estimated by the W_S/W_B ratio, is similar for shields (median 0.16) and non-shields (median 0.18), and somewhat larger for shield-like volcanoes (median 0.28 ; Table 1; Fig. 3i). Volcanoes with summit calderas tend to have higher W_S/W_B ratios (median W_S/W_B of 0.33 considering all volcanoes) (Fig. 3i). Summit calderas (with diameters $> 1.6 \text{ km}$) are much more common on shield and shield-like volcanoes (18% and 50% have a summit caldera, respectively) than on non-shield volcanoes (only 5% have a summit caldera). Summit calderas also tend to be larger on shield and shield-like volcanoes (median caldera widths of 4.6 and 5.2 km , respectively) than on non-shield volcanoes (median caldera width of 2.3 km) (Table 1; Fig. 3j).

The values of ellipticity and irregularity indexes are similar for shields, shield-like and non-shield volcanoes. Most volcanoes have $El_{AVG} < 2.5$ and $Ir_{AVG} < 1.5$ (Table 1; Fig. 3k). While both shields and non-shields have several examples with very high elongation, only one shield, Upolu, has very high irregularity ($Ir_{AVG} = 2.25$), as opposed to several non-shields.

Peak number ranges from 1 to over 200 and correlates positively with volcano basal area (Fig. 3l). Peak density (i.e. the number of peaks per km^2) is lower for shield and shield-like volcanoes (median of 0.06 and 0.05 peaks/ km^2 , respectively) than for non-shield volcanoes (median of 0.11 peaks/ km^2) (Table 1; Fig. 3l). Non-shield volcanoes with many peaks tend to have higher Ir_{AVG} values, whereas shields and shield-like volcanoes with many peaks tend to be larger.

4.1. Principal component analysis of the morphometric parameters

Table 2 shows the PCA result using the database subset of 130 volcanoes and 19 parameters. The first 4 principal components (PC) have eigenvalues > 1 , summarize 85% of the variability, and each input morphometric parameter shows a high loading on at least one of these 4 components. Therefore, only these first 4 components were retained for further analysis and are described hereafter. Each component can be correlated with a key morphological descriptor as follows:

- **Size (PC1)**: the 8 parameters describing the size of the volcano base, its height and its volume are all strongly correlated and have high loads on PC1. Summit region size and the number of peaks also contribute to PC1.
- **Steepness (PC2)**: the H/W_B ratio and the average slope values are strongly correlated and are descriptors of edifice steepness. These parameters have high loading on PC2.
- **Plan shape (PC3)**: the four ellipticity and irregularity indexes ($El_{OUTLINE}$, $Ir_{OUTLINE}$, El_{AVG} and Ir_{AVG}) have high scores on PC3. Summit region size and the W_S/W_B ratio have moderate negative loading on this PC.
- **Truncation (PC4)**: the W_S/W_B ratio, a descriptor of edifice truncation, has the highest score on PC4. Summit region size also contributes to PC4. Shields with a summit caldera tend to have higher scores in this component.

4.2. Cluster analysis, classification and description of shield types

In order to classify the volcanoes into morphometric groups we performed hierarchical clustering using the four principal components described above and the subset of 130 volcanoes. We applied three clustering methods as described in Section 3.5. Each method was applied twice, first considering the original dataset and then truncating values above a threshold defined as $Q75 + 1.5 * IQR$, where $Q75$ is the 75th percentile and IQR is the interquartile range, so as to decrease the influence of outliers. Fig. 4 shows two clustering results. Each method produces different ‘trees’ and linkage of groups, but most volcanoes tend to cluster in the same groups. Evaluating all of the results, we

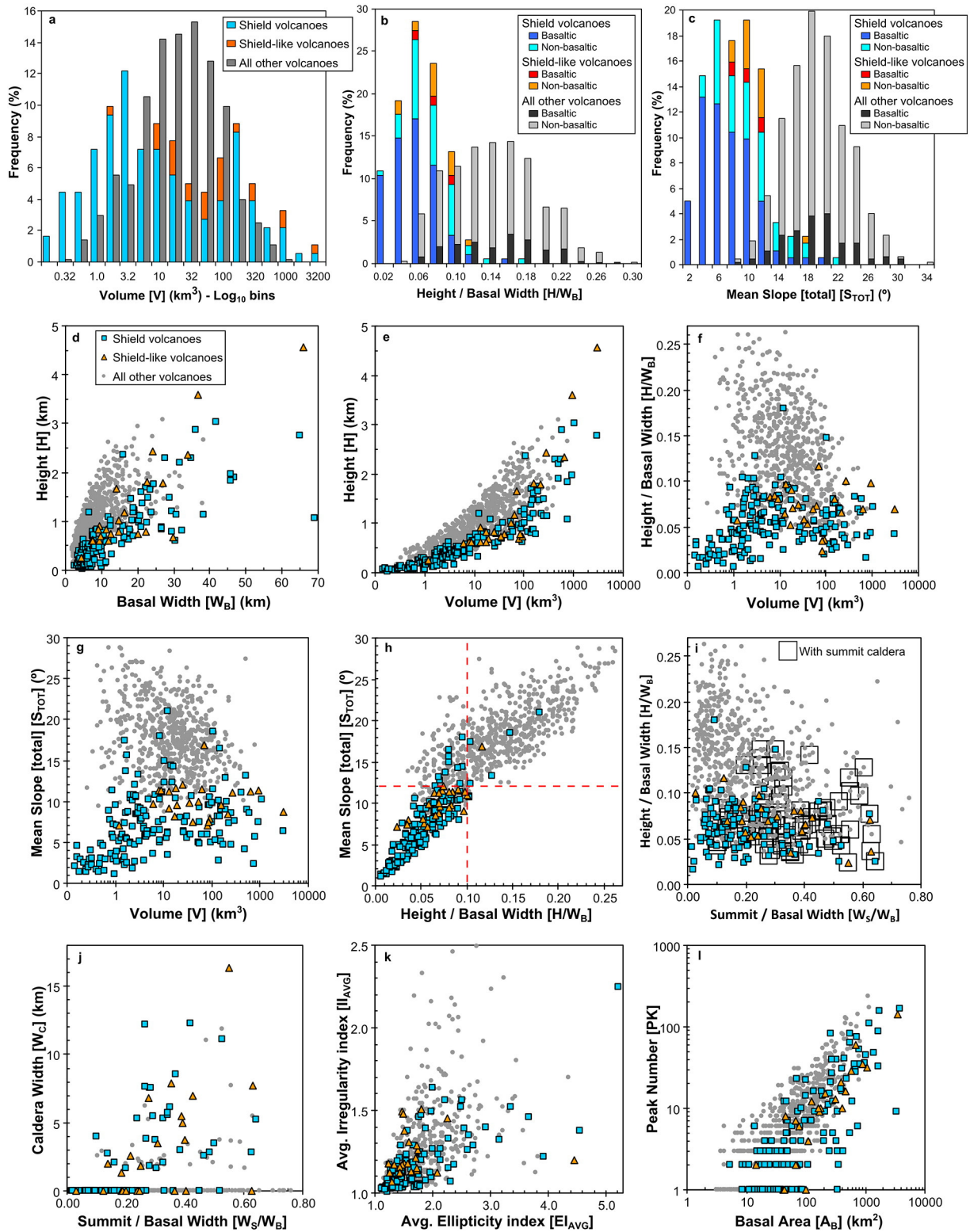


Fig. 3. Histograms and X-Y plots of morphometric parameters distinguishing between shield volcanoes, shield-like stratovolcanoes and all other stratovolcanoes/complex volcanoes. (From the database of Grosse et al. (2014).)

find that 93 volcanoes, out of the 130 included in the cluster analysis, consistently cluster into 12 groups that can be considered 12 shield types (see Electronic supplement Table 1 for details). The different methods link these 12 groups with contrasting hierarchical connections (Fig. 4); hence, a stable hierarchy of groups cannot be defined. The remaining 37 unclassified volcanoes have either intermediate values between defined clusters and cluster in different groups depending on

the applied method (i.e. they are unstable, Fig. 4), or are outliers, with extreme values in one or more components. Table 3 summarizes the main morphometric characteristics of each type (numbered from 1 to 12 in order of increasing volume), Fig. 5a plots the different clusters in PC space and Fig. 5b compares the normalized average values of the four PCs in spider diagrams. Fig. 6 shows an example of each type.

Table 2

Result of principal component analysis using the database subset of 130 volcanoes and 19 parameters.

Parameter (unit)	Abbreviation	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10
Basal area (km ²)	A _B	0.93	−0.10	−0.01	0.23	0.00	−0.19	−0.08	−0.13	0.04	0.05
Basal width (km)	W _B	0.97	−0.09	−0.02	0.06	0.08	−0.02	0.08	−0.05	−0.05	0.08
Minor basal axis (km)	mAx _B	0.95	−0.09	−0.15	0.08	0.15	−0.03	0.09	−0.02	−0.05	0.08
Major basal axis (km)	MAx _B	0.97	−0.10	0.11	0.02	0.03	0.01	0.05	−0.11	−0.06	0.03
Summit area (km ²)	A _S	0.60	−0.07	−0.53	−0.47	0.04	0.15	0.07	0.01	0.32	−0.02
Summit width (km)	W _S	0.65	−0.04	−0.53	−0.51	0.00	0.15	0.05	0.04	0.05	0.00
Height (km)	H	0.83	0.39	−0.12	0.14	0.04	0.20	0.17	0.09	−0.15	−0.05
Maximum height (km)	H _{MAX}	0.81	0.41	−0.15	0.20	−0.01	0.16	0.14	0.11	−0.15	−0.03
Volume (km ³)	V	0.84	0.08	−0.16	0.35	−0.21	−0.03	−0.26	0.06	0.07	−0.05
Maximum volume (km ³)	V _{MAX}	0.79	0.07	−0.19	0.40	−0.26	−0.06	−0.30	0.09	0.07	−0.05
Height/basal width	H/W _B	−0.21	0.91	−0.01	0.19	0.10	0.01	0.09	−0.08	0.09	−0.09
Summit width/basal width	W _S /W _B	0.03	0.02	−0.57	−0.71	−0.14	−0.15	−0.22	−0.10	−0.24	0.03
Ellipticity index [outline]	El _{OUTLINE}	0.32	−0.04	0.68	−0.28	−0.39	0.33	−0.03	−0.27	−0.02	−0.13
Ellipticity index [avg. contours]	El _{AVG}	0.42	0.07	0.76	−0.22	0.07	0.18	−0.17	0.17	0.04	0.27
Irregularity index [outline]	Il _{OUTLINE}	0.50	0.10	0.42	−0.33	−0.42	−0.37	0.32	0.20	0.04	−0.02
Irregularity index [avg. contours]	Il _{AVG}	0.52	0.03	0.48	−0.43	0.39	−0.09	−0.21	0.22	−0.03	−0.24
Mean slope [total] (°)	S _{TOT}	−0.07	0.97	0.03	−0.17	0.01	−0.06	−0.08	−0.06	0.03	0.06
Mean slope [flank] (°)	S _{FL}	−0.08	0.96	0.03	−0.17	0.00	−0.07	−0.08	−0.08	0.03	0.07
Number of peaks	PK	0.77	−0.14	0.33	−0.04	0.27	−0.20	0.05	−0.32	0.05	−0.03
Eigenvalues		8.57	3.10	2.57	1.91	0.73	0.51	0.48	0.39	0.25	0.19
Variances		0.45	0.16	0.14	0.10	0.04	0.03	0.03	0.02	0.01	0.01

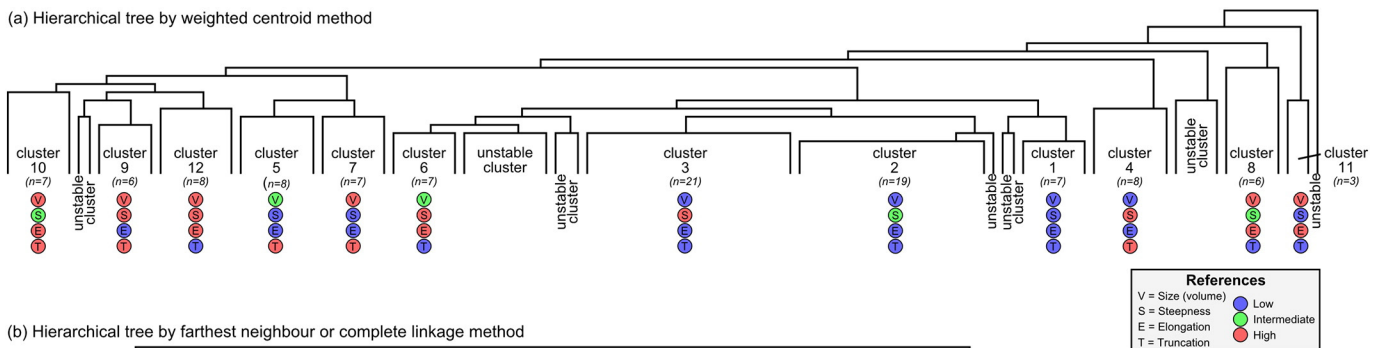
For description of parameters see Electronic supplement Table 2. Main scores of the first four principal components are highlighted in bold.

Type 1 shields are very small (<2 km³), very flat ($H/W_B < 0.04$) and sub-circular. They have smooth flanks with small summit regions, commonly topped by a pit crater or a pyroclastic cone. Type 2 shields are small and have intermediate steepness (H/W_B between 0.04 and 0.07). Although some are slightly elongated due to secondary flank vents, they are mostly sub-circular. Several are capped by small craters and/or pyroclastic cones, but none are truncated by large calderas. Type 3 shields are similar to type 2 shields in size, plan shape and lack of truncation, but are steeper ($H/W_B > 0.07$). Type 4 shields are also small, steep and sub-circular, but they are truncated by relatively large summit calderas. Type 5 shields are small to medium in size, flat, sub-circular and truncated by summit calderas. Type 6 shields are also small to medium in size, but they are steep and elongated; some have summit craters or calderas, but these are relatively small and hence they have low truncation values. Type 7 shields are large, flat, sub-circular and

truncated by large calderas. Type 8 shields are large and have variable steepness; their diagnostic characteristic is that they are strongly elongated. They have relatively small summit regions and none have summit calderas. Type 9 shields are the steep equivalent to type 7: large, steep, sub-circular and truncated by a summit caldera. Type 10 is the only group of shields that are both elongated and truncated by large summit calderas. They are large and have intermediate steepness. Type 11 consists of only 3 shields that are large to very large, flat, elongated and with very small summit regions (i.e. pointy). Type 12 shields are very large, steep and weakly to strongly elongated. Truncation is low to intermediate, and only one volcano has a summit caldera.

Of the 23 analyzed shield-like volcanoes, 15 classify into a cluster and 8 do not. The 15 classified volcanoes do not cluster into one or a few groups, but are rather spread out in most groups, with representatives in 9 of the 12 clusters (see Table 3).

(a) Hierarchical tree by weighted centroid method



(b) Hierarchical tree by farthest neighbour or complete linkage method

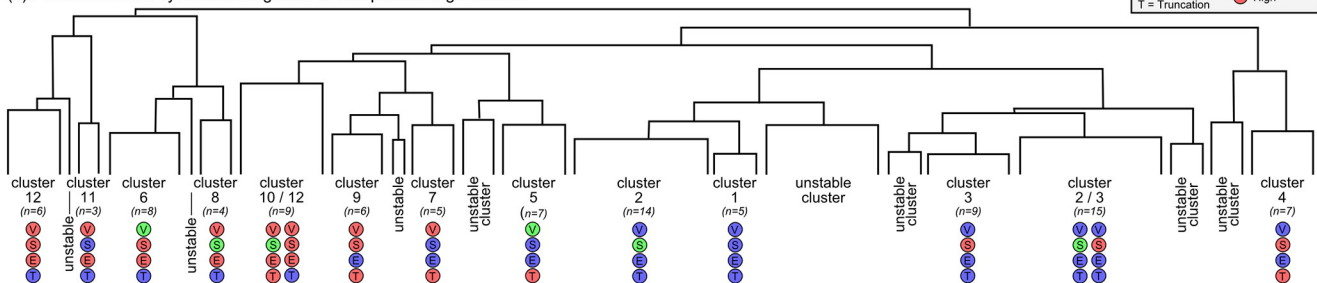
**Fig. 4.** Examples of two hierarchical trees showing linkages and clusters, and the main morphometric characteristics of each cluster; (a) weighted centroid method; (b) farthest neighbour or complete linkage method; both with truncation of extreme values.

Table 3

Summary statistics of the morphometry of the 12 shield volcano types as determined by cluster analysis. Also shown are the dominant compositions and tectonic settings for each shield type.

Cluster/type	Num. entries	Basal width	Height	Volume	Height/basal width	Summit width/basal width	Ellipticity index [avg. contours]	Irregularity index [avg. contours]	Mean slope [total]	Peak number	Calderas	Shield-like entries	Manual additions ^a	Dominant rock types ^b	Tectonic settings ^b	Crust type ^b	Regions/examples
		W _B (km)	H (km)	V (km ³)	H/W _B	W _S /W _B	El _{AVG}	Il _{AVG}	S _{TOT} (°)	PK							
1	5																
Mean		5.8	0.16	1.1	0.028	0.15	1.38	1.08	2.7	1.0	0 (0%)	0	28	Basalt	Intraplate, rift	Continental, oceanic > intermediate	Haruj VF, Ithnayn VF, Snake River Plain, Iceland, EARS
Min.		4.2	0.10	0.3	0.021	0.06	1.22	1.05	2.2	1	–						
Max.		7.0	0.23	1.8	0.035	0.20	1.49	1.12	3.2	1	–						
2	17																
Mean		6.9	0.39	4.5	0.057	0.14	1.39	1.05	6.3	1.4	0 (0%)	1	16	Basalt > andesite	Subduction/ back-arc > rift	Continental > oceanic	Kamchatka, Michoacán-Guanajuato VF, Iceland, Terevaka (Easter Is.)
Min.		3.9	0.23	0.8	0.045	0.05	1.13	1.03	4.9	1	–						
Max.		10.6	0.71	12	0.071	0.29	1.68	1.08	7.7	3	–						
3	18																
Mean		6.6	0.59	6.7	0.090	0.12	1.50	1.08	10.2	2.1	0 (0%)	3	4	Andesite > basalt	Subduction/ back-arc > rift	Continental > oceanic	Kamchatka, Michoacán-Guanajuato VF, Afderá, Ile aux Cochons
Min.		3.3	0.31	0.9	0.073	0.02	1.23	1.03	8.0	1	–						
Max.		11.8	1.08	22	0.104	0.23	1.89	1.15	12.4	7	–						
4	7																
Mean		7.4	0.56	9.9	0.076	0.44	1.49	1.16	12.9	3.1	5 (71%)	3	3	Rhyolite, basalt > andesite, dacite	Rift, subduction	Oceanic, continental	EARS, Mayor Is., Billy Mitchel, Apoyeque, Terceira, Bouvet
Min.		4.5	0.34	1.7	0.065	0.31	1.36	1.07	11.2	1	3.1						
Max.		11.3	0.83	25	0.091	0.63	1.73	1.24	15.6	8	2.6						
5	8																
Mean		12	0.48	19	0.040	0.43	1.35	1.13	6.3	7.5	7 (88%)	1	5	Basalt > trachyte, andesite, rhyolite	Rift > subduction	Continental > oceanic, intermediate	EARS, Niuao'ou, Yunaska, Marchena, Sete Cidades
Min.		6.9	0.21	2.1	0.025	0.33	1.20	1.07	3.5	1	4.3						
Max.		17.6	0.74	36	0.058	0.64	1.52	1.24	9.8	16	2.1						
6	7																
Mean		11	0.82	34	0.072	0.15	1.97	1.29	10.4	11	2 (29%)	1	1	Basalt, andesite, trachyte, rhyolite	Rift > subduction	Oceanic, continental	Pantelleira, Marion Is., Socorro, Zapatera, Fayal
Min.		7.8	0.48	6.9	0.059	0.03	1.65	1.20	9.8	3	1.8						
Max.		18.8	1.17	101	0.084	0.24	2.56	1.39	11.0	27	1.6						
7	7																
Mean		25	0.85	124	0.034	0.43	1.35	1.18	6.7	25	7	3	2	Basalt, andesite, trachyte	Subduction/ back-arc, intraplate, rift	Oceanic, intermediate, continental	Gaua, Ksudach, Okmok, Newberry, Tarso Voon, Western Galapagos (2)
Min.		20.0	0.68	81	0.023	0.28	1.14	1.03	4.6	5	(100%)						
Max.		30.8	1.14	249	0.042	0.63	1.66	1.44	9.6	64	9.5						
8	5																
Mean		24	1.22	153	0.057	0.11	4.06	1.61	10.8	59	0 (0%)	0	6	Basalt, trachybasalt > andesite	Intraplate, subduction > rift	Oceanic > continental	Upolu, Taveuni, Kirishima, San Martín, Hierro
Min.		18.2	1.02	55	0.030	0.04	2.38	1.38	7.6	21	–						
Max.		38.2	1.46	300	0.080	0.18	5.23	2.25	16.4	109	–						
9	6																
Mean		22	1.54	156	0.070	0.31	1.30	1.09	9.0	6.0	6	1	1	Basalt > andesite	Rift > subduction, intraplate	Oceanic > continental	Western Galapagos (4), San Carlos, Karkar
Min.		20.7	1.11	132	0.054	0.24	1.14	1.02	8.1	1	(100%)						
Max.		23.5	1.81	198	0.080	0.39	1.50	1.36	10.1	17	5.2						
10	4																
Mean		27	1.49	225	0.056	0.35	1.86	1.35	8.6	31	3 (75%)	1	0	Basalt > trachybasalt	Subduction	Continental, intermediate	Aoba, Ambrym, Westdahl, Payún Matrú
Min.		22.9	1.20	209	0.041	0.26	1.76	1.12	7.3	28	7.9						
Max.		29.5	1.79	250	0.066	0.42	1.95	1.59	10.0	35	3.8						
11	3																
Mean		49	1.26	465	0.027	0.02	2.29	1.32	3.1	119	0 (0%)	0	2	Basalt	Rift, intraplate	Oceanic, continental	Marsabit, Halla, Santa Cruz
Min.		32.2	0.81	141	0.015	0.01	2.00	1.17	2.4	35	–						
Max.		69.2	1.89	766	0.040	0.02	2.50	1.52	4.2	166	–						
12	6																
Mean		34	2.64	624	0.078	0.17	1.96	1.23	10.5	33	1 (17%)	1	1	Basalt > trachybasalt, andesite	Intraplate > subduction	Oceanic, continental	Mt. Cameroon, Santa Isabel, Karthala, Mauna Kea, Haleakala, Cofre de Perote
Min.		24.6	1.75	193	0.066	0.06	1.41	1.05	8.6	12	2.7						
Max.		41.9	3.59	1070	0.098	0.25	2.62	1.47	12.8	48	–						

For description of parameters see Electronic supplement Table 2.

^a Unclassified entries or those not retained in the analyzed subset that can be manually assigned to a shield type.

^b Following the Global Volcanism Program database.

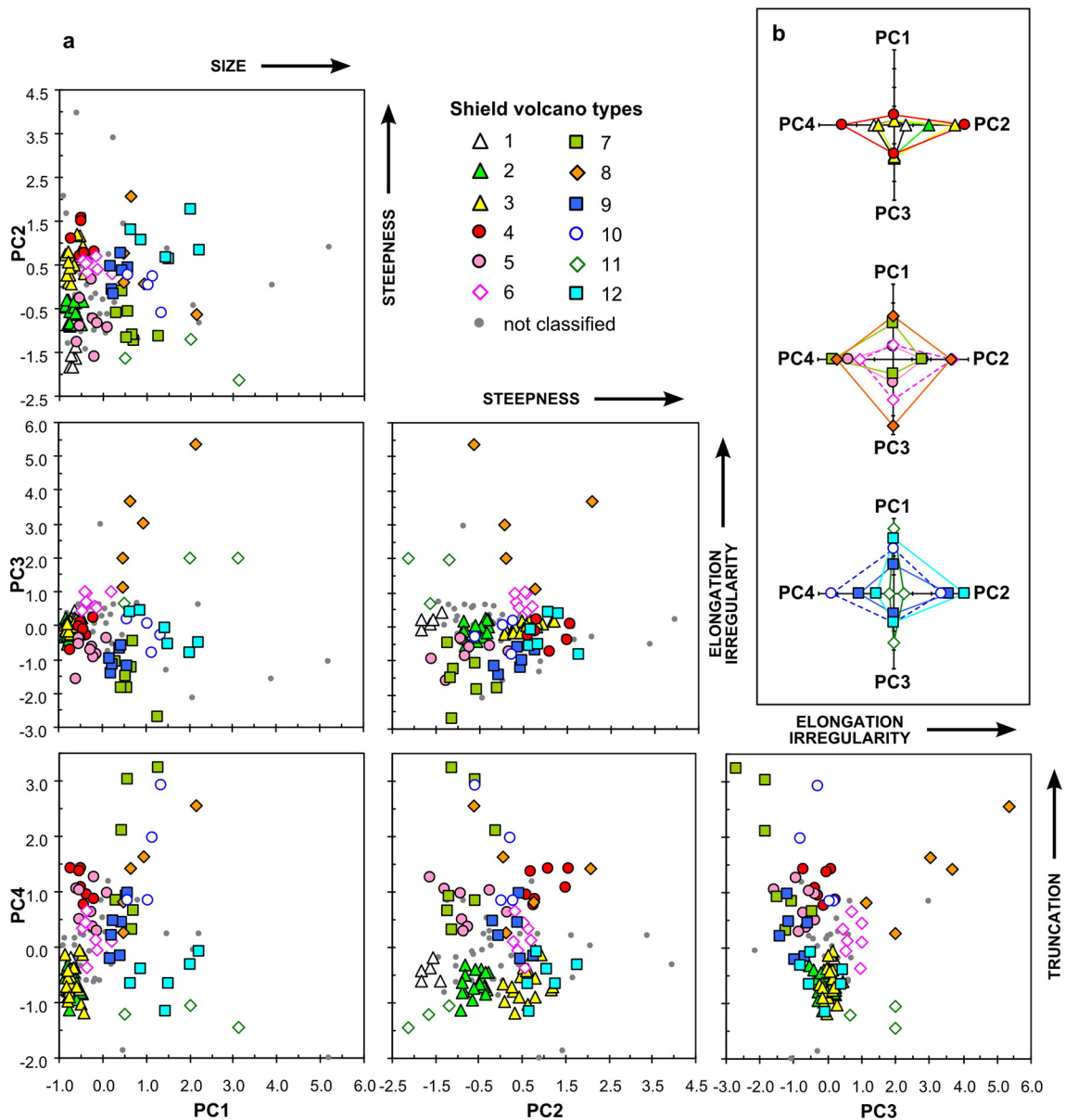


Fig. 5. a) X-Y plots of the principal components obtained from principal component analysis showing the 12 shield volcano types as determined by the cluster analysis. b) Spider diagrams of normalized average principal component values for each of the 12 shield types.

5. Discussion: shield volcano types and evolutions

The 12 obtained clusters can be regarded as end-member morphological types of shield volcanoes. Between these end-members, intermediate volcanoes remain unclassified, the whole thus defining a continuum of morphologies. Several volcanoes not retained in the analyzed subset of 130 volcanoes, as well as some unclassified volcanoes within this subset, can be assigned to a particular shield type, whereas others cannot (e.g. Etna, Piton de la Fournaise; see Electronic supplement Table 1).

Fig. 7 proposes a scheme where the main controlling factors and potential evolutionary trends linking the different shield types in terms of growth stages are highlighted. The proposed scheme emphasizes the role played by the geometry of the magmatic feeding system and the eruptive vent spatial distribution (e.g. Acocella and Neri, 2009; Tibaldi et al., 2014), either central/point source dominant, linear or scattered. The scheme also highlights the importance of summit caldera formation in the morphology of shields. Moreover, the volume range between 20 and 50 km³ with relatively few shields (Fig. 3a) is drawn as an

approximate limit between small and large shields. This size difference might be partially controlled by contrasts in the magma supply rate, but also by the longevity and localization stability of the volcanic activity.

Although only 5 volcanoes define type 1 shields, several volcanoes not in the analyzed subset (because they lack or have only 1 closed elevation contour) can be readily assigned to this group of very small and flat shields (see Electronic supplement Table 1). They are the shields in the Haruj (Libya) and Ithnayn (Saudi Arabia) fields, the Snake River Plain (USA), and most of the volcanoes from Iceland. They are all considered monogenetic and correspond to the 'low lava shield' of Pike (1978), the 'low shield' of Greeley (1982), or the 'scutulum-type shield' of Whitford-Stark (1975) (although a few larger volcanoes fall in the 'Icelandic-type' field). Some have one or two secondary vents, but they mostly seem to have grown from a single central point-source. They are all basaltic and from intraplate or rift zone settings, on both oceanic and continental crust (Table 3).

Type 2 shields correspond to the 'Icelandic-type' shield of Whitford-Stark (1975) and the 'A-type' shield of Hasenaka (1994). Type 3 shields are also mainly of the 'Icelandic-type' in terms of size, but are unusually

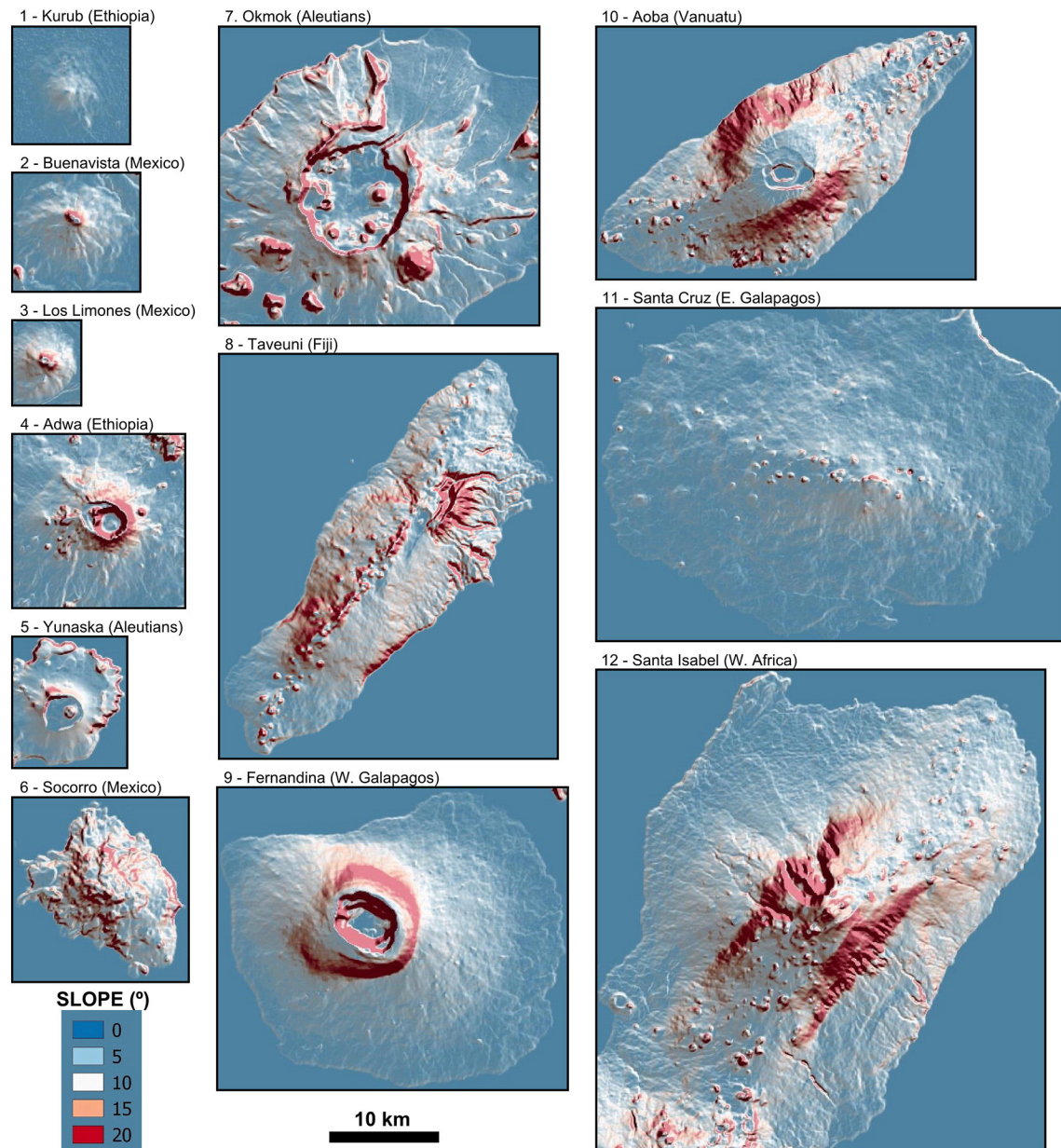


Fig. 6. Slope maps of an example for each of the 12 shield volcano types as determined by the cluster analysis. Same scale for all examples.

steep, corresponding to the 'B-type' of Hasenaka (1994). In both of these types, some volcanoes are slightly elongated due to secondary vents, but in most cases they are sub-circular and a main central point source is evident. Three volcanoes from Iceland (including famous Skjaldbreidur) are of type 2, suggesting that type 1 shields can possibly grow into type 2 shields, with steepening (Fig. 7). Rossi (1996) attributed this evolution of Icelandic shields to the relative contribution of lava lake overflow and lava tubes feeding the apron, as well as the stability of lava tubes, which, when unstable, feed multiple rootless vents on the flank of the shield and contribute to its growth in height. However, most shields of types 2 and 3 are from continental subduction to back-arc settings, and many are andesitic (Table 3), most volcanoes being from Kamchatka and the Michoacán-Guanajuato Volcanic Field (MGVF) in Mexico. These shields were probably steeper than type 1 shields from the start, because of higher viscosity and/or lower effusion rates. Although we did not analyze any steeper shields as small as the smallest of the type 1 shields (because they have basal widths below the 3-km threshold), they were visually recognized in both Kamchatka and the MGVF (Fig. 7), and Hasenaka

(1994) measured them in the MGVF. Type 3 shields are slightly more voluminous than type 2 shields, although there is much overlap. It is unclear whether there is a sharp morphological contrast between types 2 and 3, and whether type 2 shields may grow into type 3 shields (Fig. 7). The Kamchatka shields suggest a continuous range in steepness between types 2 and 3, whereas the MGVF shields suggest two distinct types, as proposed by Hasenaka (1994).

Types 1, 2 and 3 consist of many small, pointy/not truncated, point-source shields. In strong contrast, there is no equivalent cluster of large shields (i.e. large, sub-circular and pointy/not truncated). In other words, clusters of large shields are all either elongated and/or truncated. Only a few unclassified volcanoes fit the criteria of large, sub-circular and pointy, e.g. Tendürek Dagi (Turkey), which has a small 1 km-wide summit crater, Dabbahu (East Africa Rift System) and Cerro Grande in the MGVF (Fig. 7). These shields are similar to type 2 shields but larger. The existence of Cerro Grande in the MGVF suggests that type 2 shields can grow into larger shields maintaining their shape (Fig. 7). However, the scarcity of these shields indicates that the small types 1, 2 and 3

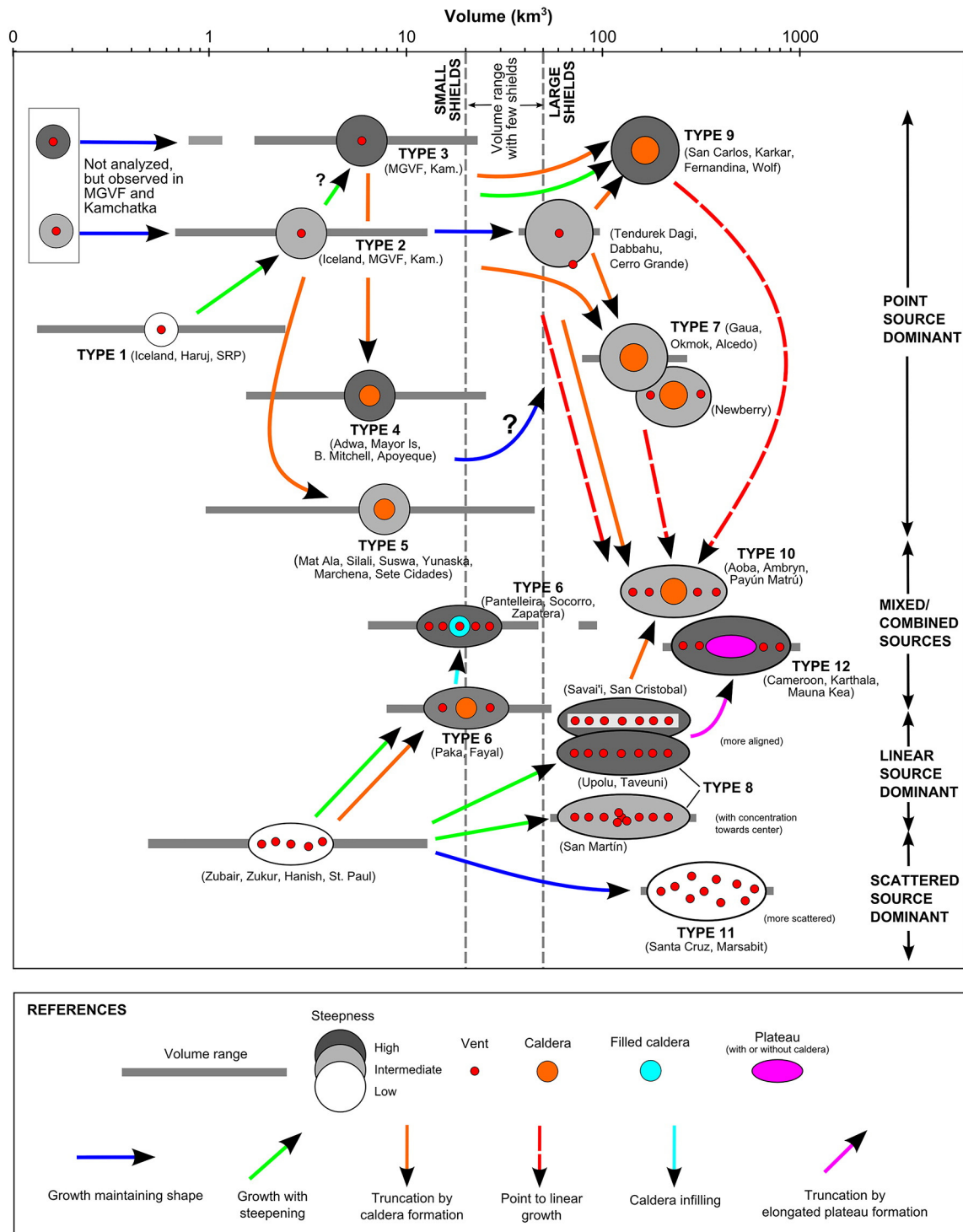


Fig. 7. Evolution scheme for shield volcanoes showing shield types (and examples) and their possible links in terms of growth stages. MGVF: Michoacán-Guanajuato Volcanic Field; Kam.: Kamchatka; SRP: Snake River Plain.

shields either stop growing (i.e. have short life spans) and/or grow very slowly due to low magma supply rate, or tend to change shape with growth, either by formation of a summit caldera (and consequent truncation) or by vent migration (and consequent elongation).

Shield types 4 and 5 are similar in that they are mostly small, sub-circular and truncated. The main difference between them is steepness. They both possibly develop from types 2 or 3 shields by caldera formation (Fig. 7). They are found on most tectonic settings and have varying compositions (Table 3). Several are from the East Africa Rift

System (6) and several are islands (7). Interestingly, two type 4 shields (Billy Mitchell, SW Pacific, and Apoyeque, Central America), as well as similar but unclassified Loloru (SW Pacific) and Tambo Quemado (Central Andes), are all classified as pyroclastic shields by the GVP.

As opposed to the previous case where an equivalent cluster of large shields was lacking, there are two clusters of large, sub-circular and truncated shields: flatter type 7 and steeper type 9 (Fig. 7). Type 7 includes the two flatter Western Galápagos shields (Alcedo and Sierra Negra), Okmok (Aleutians), and interestingly three 'stratovolcanoes'

(Tarso Voon, Chad; Gaua, SW Pacific; Ksudach, Kamchatka). Also in this group is Newberry (NW USA), which is distinct in that it has a certain elongation due to aligned secondary vents. Some unclassified volcanoes can be assigned to this type, such as Medicine Lake (NW USA) and Emi Koussi (Chad). Type 9 includes the four steeper Western Galápagos shields (Fernandina, Wolf, Darwin and Cerro Azul), San Carlos (West Africa) and Karkar (Papua New Guinea). They can be considered the 'Galápagos-type' *sensu stricto* as first described by Whitford-Stark (1975). Some unclassified volcanoes are similar to this type, but with some slight difference; for example, Tambora (Indonesia) is similar but larger, and Nyamuragira (East Africa Rift System) is similar but has a relatively small caldera and hence a lower truncation. The shields of both types 7 and 9 show a few or many secondary vents, but these are generally not abundant enough and/or are not sufficiently aligned to generate a strong elongation. In the case of the Western Galápagos shields, vents show both concentric and radial distribution patterns (Chadwick and Dietrich, 1995). Most type 7 and 9 shields are located on oceanic (e.g. Western Galápagos) or intermediate crust (e.g. Gaua, Karkar, Okmok) and are basaltic, whereas a few are from continental subduction/back-arc settings and andesitic (Ksudach, Newberry). A particular case is the two trachytic and pyroclastic-dominated volcanoes of North Africa (Tarso Voon, Emi Koussi).

The lack of large point-source dominant shields without a caldera as opposed to the abundance of those with a caldera suggests that point-source shields tend to develop a summit caldera at a certain stage in their growth (Fig. 7). This can be accounted for by the fact that as a volcano grows, it will generate more stress on the underlying crust, which will tend to trap magma into shallow magma chambers (Pinel and Jaupart, 2000). This will particularly be the case for magmatic systems with high magma supply rates capable of developing large-volume shallow magma reservoirs, in turn capable of feeding large-volume eruptions, favouring the formation of calderas (i.e. types 7 and 9). Conversely, at short-lived volcanoes, or volcanoes with low magma supply rates, large reservoirs and calderas generally do not develop (i.e. types 1, 2 and 3). Types 4 and 5 are slightly larger than types 1–3 and might represent the smaller end-member of volcanic structures capable of developing shallow magma reservoirs. Types 4 and 5, if long-lived, could eventually grow to types 7 and 9 (Fig. 7), although this would possibly require a complex evolution including caldera infilling, an increase in magma supply rate and the development of a new, larger caldera.

The abundance of shields of types 4 and 5 compared to the absence of shields of types 7 and 9 in the East Africa Rift System suggests that in continental rifts caldera formation occurs at rather lower shield volumes (and possibly at relatively lower magma supply rates). However, in back-arc (Newberry, Ksudach) and intraplate (Tarso Voon, Emi Koussi) continental settings large type 7 and 9 shields can develop. In oceanic settings caldera formation can occur at both small and large volumes, as suggested by several oceanic shields in both types 4/5 and 7/9, although it should be kept in mind that some 'small' oceanic shields of types 4 and 5 continue underwater and hence are actually much larger.

Shields of type 6 are steep and elongated, due to aligned secondary vents. Although they are rather pointy, some (e.g. Paka, East Africa Rift System; Fayal, Azores) have small summit calderas, whereas others (e.g. Pantelleira, Italy; Socorro, Mexico; Zapatera, Central America) are characterized by an ancient large caldera partially or completely masked by the growth of a new edifice within it. Hence, type 6 shields seem to have elements of both point and linear sources superposed (Fig. 7). This shield type occurs both in oceanic and continental rift settings and their compositions are highly variable (Table 3).

Shield types 8 and 11 are large, elongated shields lacking truncation/summit calderas. They have variable steepness values and, together with several unclassified volcanoes (e.g. Karaca Dag, Turkey; Savai'i, SW Pacific; Hualalai, Hawaii; Santiago and San Cristóbal, Eastern Galápagos) form a continuum from very flat to steep edifices. The main characteristic of these shields is their high elongation due to the abundance of aligned or sub-aligned vents reflecting a dominant linear

or spatially distributed magmatic source. Most have well developed volcanic rift zones, and the vents show varying degrees of alignment, from very well aligned to quite scattered. The flattest volcanoes (e.g. Marsabit and Tepi, East Africa Rift System; Santa Cruz, Eastern Galápagos), resemble the morphology of cone fields rather than of an individual edifice, with a scattered distribution of vents, but yet with higher elevation in the center. Those with higher H/W_B ratios and average slope values tend to have steeper upper flanks, usually in the form of an elongated ridge, with either a linear spine-like summit (e.g. Upolu and Taveuni, SW Pacific; Cumbre Vieja, Canary Islands), or with a wider, flatter elongated summit area (e.g. San Cristóbal; Savai'i), or with more sub-circular upper flanks and a pointy summit (e.g. Halla, South Korea; San Martín, Mexico) (Fig. 7). Mauna Loa (Hawaii) can be considered an outlier of this group because of its extremely large size. Most of these large elongated shields are of basaltic composition, whereas a few are trachybasaltic and only one is andesitic (Kirishima, Japan). They are located mostly on oceanic hotspots/rifts (e.g. Hawaii, Eastern Galápagos, Canary Islands), but also on continental rifts (e.g. Marsabit) and subduction zones (e.g. SW Pacific).

No clusters of small shields equivalent to types 8 and 11 were obtained in the cluster analysis. However, several volcanoes that were not analyzed (because of few or no closed contours) or classified have similar elongated shapes, with aligned or sub-aligned vents (e.g. Zukur and Hanish, East Africa Rift System; St. Paul, West Alaska). These small shields are usually very flat and form small cone fields; they can be considered precursors to the large type 8 and 11 shields (Fig. 7). With growth, they can maintain the cone-field appearance, usually with a higher density of cones in the center, and grow into the flatter elongated shields (e.g. Marsabit), or, more commonly, they become steeper with growth, forming linear ridges (e.g. Taveuni, Hualalai) or a more circular central area (e.g. San Martín, Halla) (Fig. 7).

Type 10 shields are large, truncated and have intermediate steepness, whereas type 12 shields are larger, steeper and have a less pronounced truncation. Both types of shields are elongated, due to the alignment of vents along rift zones, and have summit calderas (type 10) and/or flat, circular to somewhat elongated, summit regions or plateaus (type 12). Hence, these shield types seem to combine linear and point sources (Fig. 7). Some (Mauna Kea and Haleakala, Hawaii; Ambrym, SW Pacific) are characterized by well-formed triple rift zone structures with an established central magmatic system (e.g. Moore and Clague, 1992; Walker, 1999). Others (Karthala, Indian Ocean; Aoba, SW Pacific; and to a lesser extent Mount Cameroon and Santa Isabel, West Africa) show a combination of a single linear rift zone coupled with central growth, leading to steep flanks perpendicular to the rift zone in the central part of the edifice and formation of a large summit plateau (e.g. Annen et al., 2001). The latter may be attributed to subsidence of the summit area due to deformation of hydrothermally altered rocks (Merle et al., 2010) or loading of the shallow intrusive system (Walker, 1988).

Shield types 10 and 12 may evolve either from dominantly point source shields (e.g. types 7 and 9) or from dominantly linear source shields (e.g. type 8) (Fig. 7). In the first case, the linear component develops from the volcano stress field as it grows larger and deforms under gravity, often coupled with a secondary regional tectonic control, with consequent formation of aligned flank vents following the local stress field (e.g. Fiske and Jackson, 1972; Walter and Troll, 2003; Kervyn et al., 2009). In the second case, regional tectonics dictates vent alignment from the start of the volcano's evolution; continuing growth may produce steepening due to the accumulation of pyroclastic material along the preferential vent alignments (Rowland and Garbeil, 2000) and/or, as with the evolution of shield types 7 and 9, the establishment of shallow magma reservoirs that will favour caldera or plateau formation. In order to develop such large-volume edifices, type 10 and 12 shields must be linked to long-lived magmatic sources with high production rates. Shield types 10 and 12 are found mostly on oceanic and intermediate crust intraplate settings (e.g. Hawaii, Karthala, Mount Cameroon), but also in oceanic and continental subduction

zones (e.g. SW Pacific, Aleutians). Basaltic-trachybasaltic compositions dominate.

6. Conclusions

Shield volcanoes have long been recognized as a specific volcano type with a distinct morphology characterized by gentle slopes. Volcanoes classified as shields are generally flatter, but many volcanoes considered as shields are quite steep, with average slopes well above the thresholds used to define shields as such (see Section 2). Conversely, several volcanoes considered as stratovolcanoes (the ‘shield-like’ volcanoes of this study) are morphometrically indistinguishable from shields, and some even classify within the flatter shield types. These discrepancies highlight the fact that volcanoes show a continuum of morphologies with no clear-cut morphometric difference between shield volcanoes and non-shield volcanoes (i.e. stratovolcanoes, complex/compound volcanoes). Keeping this in mind, an approximate steepness threshold between shield and non-shield volcanoes (as considered in the literature, e.g. the GVP database) can best be drawn at 12° S_{TOT} and 0.10 H/W_B ratio (Fig. 3). Basaltic shields tend to have lower slopes than non-basaltic shields; most shields with $S_{TOT} < 5^\circ$ are basaltic, whereas most shields with $S_{TOT} > 10^\circ$ are non-basaltic. However, even lava-dominated basaltic shields (shield volcanoes *sensu stricto*) can have slopes significantly steeper than the threshold values mentioned in the literature.

Analysis with the MORVOLC algorithm of DEMs now available for most regions of the world, combined with data aggregation and classification techniques (PCA, cluster analysis), enables to rationalize the morphological variability of shield volcanoes. Results of the PCA highlight that the morphology of shield volcanoes can be summarized and quantified with four morphometric descriptors: size, steepness (average slopes and H/W_B ratio), plan shape (mainly elongation, but also irregularity) and truncation (related to presence of a summit caldera). Hierarchical cluster analysis of these key descriptors results in 12 end-member shield types, with intermediate cases defining a continuum of morphologies. A similar approach could be used to synthesize and classify the morphologies of other types of volcanic edifices.

Although some shield volcano types occur only at specific tectonic settings and/or are characterized by a dominant rock type (see Section 5 and Table 3), there is no straightforward link between the shield types, tectonic setting and composition. The shield volcano types can be linked in terms of growth stages and shape evolution depending on three main groups of factors: (1) the magma composition, rheology and effusion rate, responsible for the proportion of lava vs. pyroclast output and the length/thickness ratio of lava flows, which will condition edifice steepness; (2) the spatial distribution of vents and the magma output at each vent, dependent on the magmatic feeding system that is in turn controlled by both the initial regional tectonic framework and the subsequent local stress field induced by the edifice load, which will control edifice plan shape as well as steepness; and (3) summit caldera formation, which will increase edifice truncation.

The interplay of these factors at different stages in shield volcano evolution will result in the different observed morphologies. The shape of small shields with a simple, point-source feeding system and formed over short time-spans and/or having low magma supply rates (types 1, 2 and 3) will be mainly controlled by the magma composition and rheology and the associated length/thickness ratio of the lava flows. Some of these small shields may develop a summit caldera (types 4 and 5), but caldera formation mostly occurs at larger shields. The longevity of the magmatic source and the magma supply rate probably control the capacity of shields to grow to large volumes and to develop caldera-generating shallow magma reservoirs. Hence, point-source dominated, high supply rate growth over longer time periods will produce large shields with more complex shallow magmatic systems leading to the formation of summit calderas (types 7 and 9). The loading on the crust of these large shields will favour the development of

peripheric vents. Under regional tectonic stress, and with the positive feedback of the local stress induced by the volcanic edifice itself, these larger shields may develop more or less defined rift zones and grow elongated (types 10 and 12). A separate evolutionary trend arises when the magmatic system is dominantly diffuse or linear from the start. The spatial distribution of vents will be the main factor controlling the shield shape. Initial small constructs, characterized by monogenetic shield and/or scoria cone and lava flow assemblies, may develop a point-source system at an early stage (type 6) or may grow into large elongated edifices (types 8 and 11), with the different degrees of elongation and steepness dependent mainly on vent configuration, but also on magma composition and proportion of pyroclastic products. Eventually, large and elongated shields may develop a central magmatic system and a more circular summit plateau or caldera (types 10 and 12), hence linking the two evolutionary trends.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.geomorph.2017.12.017>.

Acknowledgments

We are grateful to the Belgian Science Policy Office (BELSPO) for funding a research fellowship stay of PG at the Vrije Universiteit Brussel (VUB), and to the Université Blaise Pascal for funding short research stays of both PG and MK at the Laboratoire Magmas et Volcans (Université Blaise Pascal, Clermont-Ferrand, France). We thank Ben van Wyk de Vries and Paul Byrne for helpful discussions. We thank Eric Christiansen and Laurent Michon for their detailed and thoughtful reviews.

References

- Acocella, V., Neri, M., 2009. Dike propagation in volcanic edifices: overview and possible developments. *Tectonophysics* 471, 67–77.
- Annen, C., Lénat, J.-F., Provost, A., 2001. The long-term growth of volcanic edifices: numerical modelling of dyke intrusion and lava-flow emplacement. *J. Volcanol. Geotherm. Res.* 105, 263–289.
- Barberi, F., Varet, J., 1970. The Erta Ale volcanic range (Danakil depression, Northern Afar, Ethiopia). *Bull. Volcanol.* 34, 848–917.
- Bleacher, J.E., Greeley, R., 2008. Relating volcanic morphometry to the developmental progression of Hawaiian shield volcanoes through slope and hypsometric analyses of SRTM data. *J. Geophys. Res. Solid Earth* 113, B09208. <https://doi.org/10.1029/2006JB004661>.
- Byrne, P.K., Holohan, E.P., Kervyn, M., van Wyk de Vries, B., Troll, V.R., Murray, J.B., 2013. A sagging-spreading continuum of large volcano structure. *Geology* 41, 339–342.
- Byrne, P.K., Holohan, E.P., Kervyn, M., van Wyk de Vries, B., Troll, V.R., 2015. Analogue modelling of volcano flank terrace formation on Mars. In: Platz, T., Massironi, M., Byrne, P.K., Hiesinger, H. (Eds.), *Volcanism and Tectonism Across the Inner Solar System*. Geological Society of London Special Publication vol. 401:pp. 185–202. <https://doi.org/10.1144/SP401.14>.
- Cas, R.A.F., Wright, J.V., 1988. *Volcanic Successions Modern and Ancient. A Geological Approach to Processes, Products and Successions*. Unwin Hyman, London (528 pp).
- Chadwick, W.W., Dietrich, J.H., 1995. Mechanical modeling of circumferential and radial dike intrusion on Galápagos volcanoes. *J. Volcanol. Geotherm. Res.* 66, 37–52.
- Chadwick, W.W., Howard, K.A., 1991. The pattern of circumferential and radial eruptive fissures on the volcanoes of Fernandina and Isabela Islands, Galápagos. *Bull. Volcanol.* 53, 259–275.
- Cotton, C.A., 1944. *Volcanoes as Landscape Forms*. Withcombe & Tombs, Christchurch, New Zealand (416 pp).
- Cullen, A.B., McBirney, A.R., Rogers, R.D., 1987. Structural controls on the morphology of Galápagos shields. *J. Volcanol. Geotherm. Res.* 34, 143–151.
- Davidson, J.P., De Silva, S.L., 2000. Composite Volcanoes. In: Sigurdsson, H., Houghton, B.F., McNutt, S.R., Rymer, H., Stix, J. (Eds.), *Encyclopedia of Volcanoes*. Academic Press, San Diego, pp. 663–680.
- Donnelly-Nolan, J.M., Grove, T.L., Lanphere, M.A., Champion, D.E., Ramsey, D.W., 2008. Eruptive history and tectonic setting of Medicine Lake Volcano, a large rear-arc volcano in the southern Cascades. *J. Volcanol. Geotherm. Res.* 177, 313–328.
- Euillades, L.D., Grosse, P., Euillades, P.A., 2013. NETVOLC: an algorithm for automatic delimitation of volcano edifice boundaries using DEMs. *Comput. Geosci.* 56, 151–160.
- Fiske, R.S., Jackson, E.D., 1972. Orientation and growth of Hawaiian volcanic rifts: the effect of regional structure and gravitational stresses. *Proc. R. Soc. Lond. A* 329, 299–326.
- Francis, P., Oppenheimer, C., 2003. *Volcanoes*. Oxford University Press (534 pp).
- Greeley, R., 1982. The Snake River Plain, Idaho: representative of a new category of volcanism. *J. Geophys. Res.* 87, 2705–2712.
- Grosse, P., van Wyk de Vries, B., Petrinovic, I.A., Euillades, P.A., Alvarado, G.E., 2009. Morphometry and evolution of arc volcanoes. *Geology* 37, 651–654.

- Grosse, P., van Wyk de Vries, B., Euillades, P.A., Kervyn, M., Petrinovic, I.A., 2012. Systematic morphometric characterization of volcanic edifices using digital elevation models. *Geomorphology* 136, 114–131.
- Grosse, P., Euillades, P.A., Euillades, L.D., van Wyk de Vries, B., 2014. A global database of composite volcano morphometry. *Bull. Volcanol.* 76, 784.
- Gutmann, J.T., 2007. Geologic studies in the Pinacate volcanic field. *J. Southwest* 49, 189–243.
- Hasenaka, T., 1994. Size, distribution, and magma output rate for shield volcanoes of the Michoacán-Guanajuato Volcanic Field, Central Mexico. *J. Volcanol. Geotherm. Res.* 63, 13–31.
- Hernando, I.R., Llambías, E.J., González, P.D., Sato, K., 2012. Volcanic stratigraphy and evidence of magma mixing in the Quaternary Payún Matrú volcano, andean backarc in western Argentina. *Andean Geol.* 39, 158–179.
- Higgins, M.W., 1973. Petrology of Newberry Volcano, Central Oregon. *Bull. Geol. Soc. Am.* 84, 455–488.
- Jarvis, A., Reuter, H.I., Nelson, A., Guevara, E., 2008. Hole-filled SRTM for the globe, version 4: CGIAR-CSI SRTM 90 m Database. <http://srtm.csi.cgiar.org>.
- Jolliffe, I.T., 2002. Principal component analysis. Springer Series in Statistics, second edition Springer-Verlag, New York (488 pp).
- Kervyn, M., Ernst, G.G.J., van Wyk de Vries, B., Mathieu, L., Jacobs, P., 2009. Volcano load control on dyke propagation and vent spatial distribution: insights from analogue modelling. *J. Geophys. Res. Solid Earth* 114, B03401. <https://doi.org/10.1029/2008JB005653>.
- Kervyn, M., Van Wyk de Vries, B., Walter, T.R., Njome, M.S., Suh, C.E., Ernst, G.G.J., 2014. Directional flank spreading at Mount Cameroon volcano: evidence from analogue modeling. *J. Geophys. Res. Solid Earth* 119:7542–7563. <https://doi.org/10.1002/2014JB011330>.
- Köhn, H.-F., Hubert, L.J., 2015. Hierarchical Cluster Analysis. Wiley StatsRef: Statistics Reference Online, pp. 1–13.
- Kreslavsky, M.A., Head, J.W., 1999. Morphometry of small shield volcanoes on Venus: implication for the thickness of regional plains. *J. Geophys. Res. Planets* 104: 18925–18932. <https://doi.org/10.1029/1999JE001042>.
- Kuntz, M.A., Covington, H.R., Schorr, L.J., 1992. An overview of basaltic volcanism of the eastern Snake River Plain, Idaho. In: Link, P.K., Kuntz, M.A., Platt, L.P. (Eds.), *Regional Geology of Eastern Idaho and Western Wyoming*. Geological Society of America Memoir 179, pp. 227–267.
- Lockwood, J.P., Hazlett, R.W., 2010. *Volcanoes: Global Perspectives*. Wiley-Blackwell, UK (552 pp).
- Macdonald, G.A., 1972. *Volcanoes*. Prentice-Hall, Englewood Cliffs, New Jersey, p. 510.
- Merle, O., Barde-Cabusson, S., van Wyk de Vries, B., 2010. Hydrothermal calderas. *Bull. Volcanol.* 72, 131–147.
- Michon, L., Saint-Ange, F., 2008. Morphology of Piton de la Fournaise basaltic shield volcano (La Réunion Island): characterization and implication in the volcano evolution. *J. Geophys. Res. Solid Earth* 113, B03203. <https://doi.org/10.1029/2005JB004118>.
- Michon, L., Cayol, V., Letourneur, L., Peltier, A., Villeneuve, N., Staudacher, T., 2009. Edifice growth, deformation and rift zone development in basaltic setting: insights from Piton de la Fournaise shield volcano (Réunion Island). *J. Volcanol. Geotherm. Res.* 184, 14–30.
- Miller, T.P., McGimsey, R.G., Richter, D.H., Riehle, J.R., Nye, C.J., Yount, M.E., Dumoulin, J.A., 1998. Catalog of the historically active volcanoes of Alaska. U.S. Geological Survey Open-file Report 98-582.
- Moore, J.G., Clague, D.A., 1992. Volcano growth and evolution of the island of Hawaii. *Bull. Geol. Soc. Am.* 104, 1471–1484.
- Moore, J.G., Mark, R.K., 1992. Morphology of the Island of Hawaii. *GSA Today* 2, 257–262.
- Morgan, J.K., Moore, G.F., Clague, D.A., 2003. Slope failure and volcanic spreading along the submarine south flank of Kilauea volcano, Hawaii. *J. Geophys. Res. Solid Earth* 108:2415. <https://doi.org/10.1029/2003JB002411>.
- Mouginis-Mark, P.J., Rowland, S.K., Garbeil, H., 1996. Slopes of western Galápagos volcanoes from airborne interferometric radar. *Geophys. Res. Lett.* 23, 3767–3770.
- Naumann, T., Geist, D., 2000. Physical volcanology and structural development of Cerro Azul Volcano, Isabela Island, Galápagos: implications for the development of Galápagos-type shield volcanoes. *Bull. Volcanol.* 61, 497–514.
- Nordlie, B.E., 1973. Morphology and structure of Western Galápagos volcanoes and model for their origin. *Geol. Soc. Am. Bull.* 84, 2931–2955.
- Pedersen, G.B.M., Grosse, P., 2014. Morphometry of subaerial shield volcanoes and glaciovolcanoes from Reykjanes Peninsula, Iceland: effects of eruption environment. *J. Volcanol. Geotherm. Res.* 282, 115–133.
- Pike, R.J., 1978. Volcanoes on the inner planets: some preliminary comparisons of gross topography. *Proceedings of the 9th Lunar Planetary Science Conference*, pp. 3239–3273.
- Pike, R.J., Clow, G.D., 1981. Revised classification of terrestrial volcanoes and a catalog of topographic dimensions with new results on edifice volume. U.S. Geological Survey Open-file Report OF 81-1038.
- Pinel, V., Jaupart, C., 2000. The effect of edifice load on magma ascent beneath a volcano. *Philos. Trans. R. Soc. Lond. A* 358, 1515–1532.
- Platz, T., Münn, S., Walter, T.R., Procter, J.N., McGuire, P.C., Dumke, A., Neukum, G., 2011. Vertical and lateral collapse of Tharsis Tholus, Mars. *Earth Planet. Sci. Lett.* 305, 445–455.
- Plescia, J.B., 2004. Morphometric properties of Martian volcanoes. *J. Geophys. Res. Planets* 109:E03003. <https://doi.org/10.1029/2002JE002031>.
- Rabus, B., Eineder, M., Roth, A., Bamler, R., 2003. The Shuttle Radar Topography Mission – a new class of digital elevation models acquired by spaceborne radar. *ISPRS J. Photogramm. Remote Sens.* 57, 241–262.
- Rossi, M.J., 1996. Morphology and mechanism of eruption of postglacial shield volcanoes in Iceland. *Bull. Volcanol.* 57, 530–540.
- Rowland, S.K., 1996. Slope, lava flow volumes, and vent distributions on Volcán Fernandina, Galápagos Islands. *J. Geophys. Res. Solid Earth* 101:27657–27672. <https://doi.org/10.1029/96JB02649>.
- Rowland, S.K., Garbeil, H., 2000. Slopes of oceanic basalt volcanoes. In: Mouginis-Mark, P.J., Crisp, J.A., Fink, J.H. (Eds.), *Remote Sensing of Active Volcanism*. American Geophysical Union Geophysical Monograph Series 116, pp. 223–247.
- Rowland, S.K., Munro, D.C., Perez-Oviedo, V., 1994. Volcán Ecuador, Galápagos Islands: erosion as a possible mechanism for the generation of steep-sided basaltic volcanoes. *Bull. Volcanol.* 56, 271–283.
- Siebert, L., Simkin, T., Kimberly, P., 2010. *Volcanoes of the World*. University of California Press, Berkeley (568 pp).
- Spudis, P.D., McGovern, P.J., Kiefer, W.S., 2013. Large shield volcanoes on the Moon. *J. Geophys. Res. Planets* 118:1063–1081. <https://doi.org/10.1002/jgre.20059>.
- Tibaldi, A., Bonali, F.L., Corazzato, C., 2014. The diverging volcanic rift system. *Tectonophysics* 611, 94–113.
- Walker, G.P.L., 1988. Three Hawaiian calderas: an origin through loading by shallow intrusions? *J. Geophys. Res. Solid Earth* 93:14773–14784. <https://doi.org/10.1029/JB093iB12p14773>.
- Walker, G.P.L., 1999. Volcanic rift zones and their intrusion swarms. *J. Volcanol. Geotherm. Res.* 94, 21–34.
- Walker, G.P.L., 2000. Basaltic volcanoes and volcanic systems. In: Sigurdsson, H., Houghton, B.F., McNutt, S.R., Rymer, H., Stix, J. (Eds.), *Encyclopedia of Volcanoes*. Academic Press, San Diego, pp. 283–289.
- Walter, T., Troll, V.R., 2003. Experiments on rift zone evolution in unstable volcanic edifices. *J. Volcanol. Geotherm. Res.* 127, 107–120.
- Webb, P.K., Weaver, S.D., 1975. Trachyte shield volcanoes: a new volcanic form from South Turkana, Kenya. *Bull. Volcanol.* 39, 294–312.
- Werner, S.C., 2014. Shield volcano. In: Hargitai, H., Kereszturi, A. (Eds.), *Encyclopedia of Planetary Landforms*. Springer Science, New York, pp. 1926–1929.
- Whitford-Stark, J.L., 1975. Shield volcanoes. In: Fielder, G., Wilson, L. (Eds.), *Volcanoes of Earth, Moon and Mars*. Elek Science, London, UK, pp. 66–74.
- Wolfe, E.W., Wise, W.S., Dalrymple, G.B., 1997. The geology and petrology of Mauna Kea volcano, Hawaii – a study of postshield volcanism. U.S. Geological Survey Professional Papers 1557 (129 pp).
- Wood, C.A., 1979. Monogenetic volcanoes of the terrestrial planets. *Proceedings of the 10th Lunar Planetary Science Conference*, pp. 2815–2840.
- Zimbelman, J.R., 2000. Volcanism on Mars. In: Sigurdsson, H., Houghton, B.F., McNutt, S.R., Rymer, H., Stix, J. (Eds.), *Encyclopedia of Volcanoes*. Academic Press, San Diego, pp. 771–784.